LA HW

Included exercises: 1, 4, 6, 9, 15, 61, 64, 73, 81

Exercises 1-4

In Exercises 1–4, the reduced row echelon form of a matrix A is given. Determine the dimension of (a) Col A, (b) Null A, (c) Row A, and (d) Null A^T.

Original Exercise 1

$$\begin{bmatrix} 1 & -3 & 0 & 2 \\ 0 & 0 & 1 & -4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Original Exercise 4

$$\begin{bmatrix} 1 & 0 & 0 & -4 & 2 \\ 0 & 1 & 0 & 2 & -1 \end{bmatrix}$$

Exercises 5-12

In Exercises 5–12, a matrix A is given. Determine the dimension of (a) Col A, (b) Null A, (c) Row A, and (d) Null A^T.

$$\begin{bmatrix} 1 & 2 & -3 \end{bmatrix}$$

Original Exercise 6

$$\begin{bmatrix} 1 & 2 & -3 \\ 0 & -1 & -1 \\ 1 & 4 & -1 \end{bmatrix}$$

Original Exercise 9

$$\begin{bmatrix} 1 & 1 & 2 & 1 \\ -1 & -2 & 2 & -2 \end{bmatrix}$$

Exercises 13-16

In Exercises 13–16, a subspace is given. Determine its dimension.

Original Exercise 15

$$\left\{ \begin{bmatrix} -3s + 4t \\ s - 2t \\ 2s \end{bmatrix} \in \mathcal{R}^3 : s \text{ and } t \text{ are scalars} \right\}$$

Exercises 61-68

In Exercises 61–68, use the results of this section to show that $\mathcal{B}$ is a basis for each subspace V.

Original Exercise 61

$$\mathcal{B} = \{\mathbf{e}_1, \mathbf{e}_2\},\ V = \left\{ \begin{bmatrix} 2s - t \\ s + 3t \end{bmatrix} \in \mathcal{R}^2 : s \text{ and } t \text{ are scalars} \right\}$$

Original Exercise 64

$$\mathcal{B} = \left\{ \begin{bmatrix} 0 \\ 1 \\ 4 \end{bmatrix}, \begin{bmatrix} 4 \\ -7 \\ 0 \end{bmatrix} \right\},$$

$$V = \left\{ \begin{bmatrix} -s + t \\ 2s - t \\ s + 3t \end{bmatrix} \in \mathcal{R}^3 : s \text{ and } t \text{ are scalars} \right\}$$

Exercise 73

75. Use Exercise 73 to prove that, for any $m \times n$ matrix A and any $n \times p$ matrix B, $\operatorname{rank} AB \leq \operatorname{rank} A$.

Original Exercise 73

Prove that, for any $m \times n$ matrix A and any $n \times p$ matrix B, the column space of AB is contained in the column

Original Exercise 81

and W.

Let V be a k-dimensional subspace of $\mathcal{R}^n$ having a basis $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_k\}$. Define a function $T\colon \mathcal{R}^k \to \mathcal{R}^n$ by

$$T\left(\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_k \end{bmatrix}\right) = x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + \cdots + x_k\mathbf{u}_k.$$

(a) Prove that T is a linear transformation.

(b) Prove that T is one-to-one.